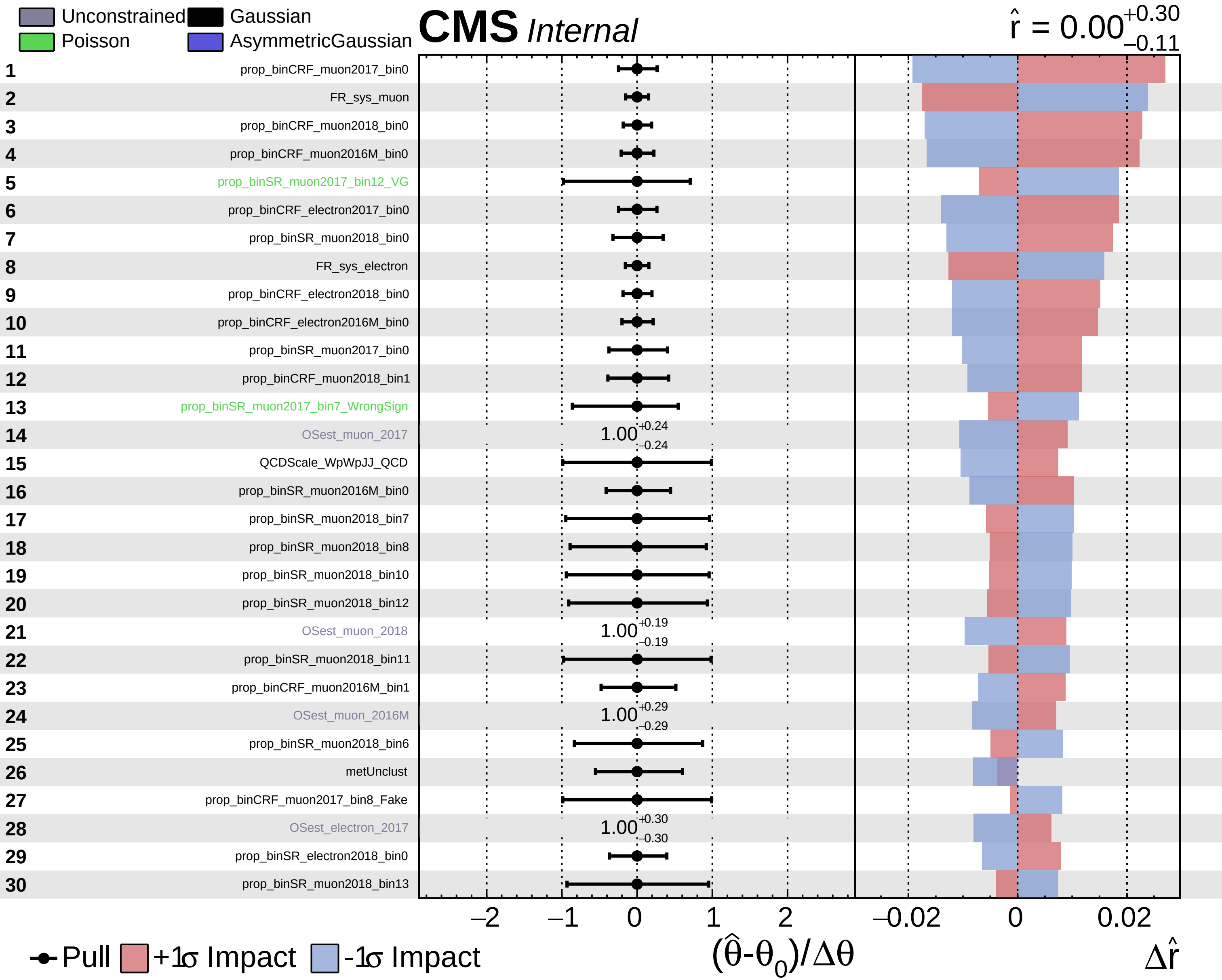
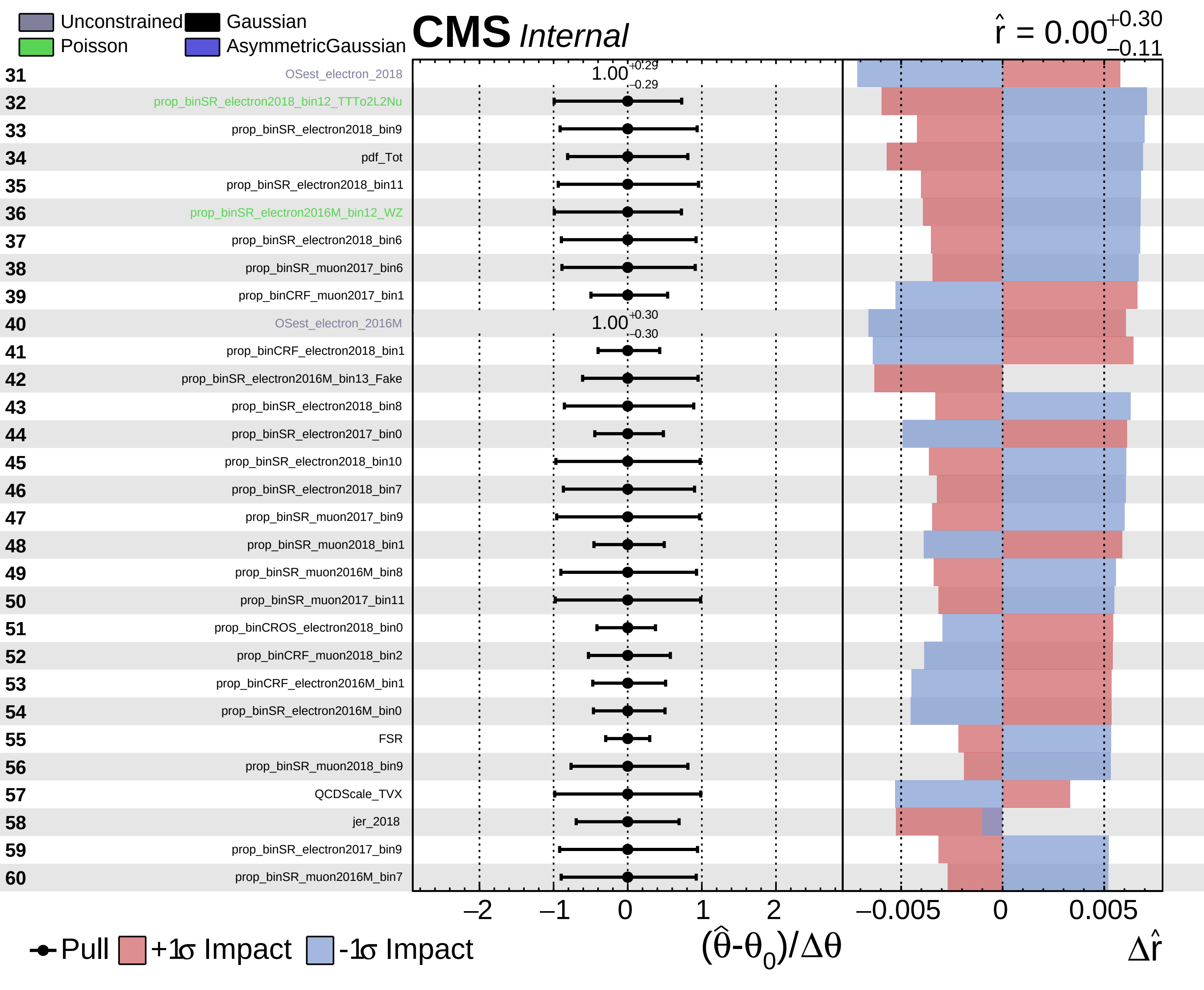
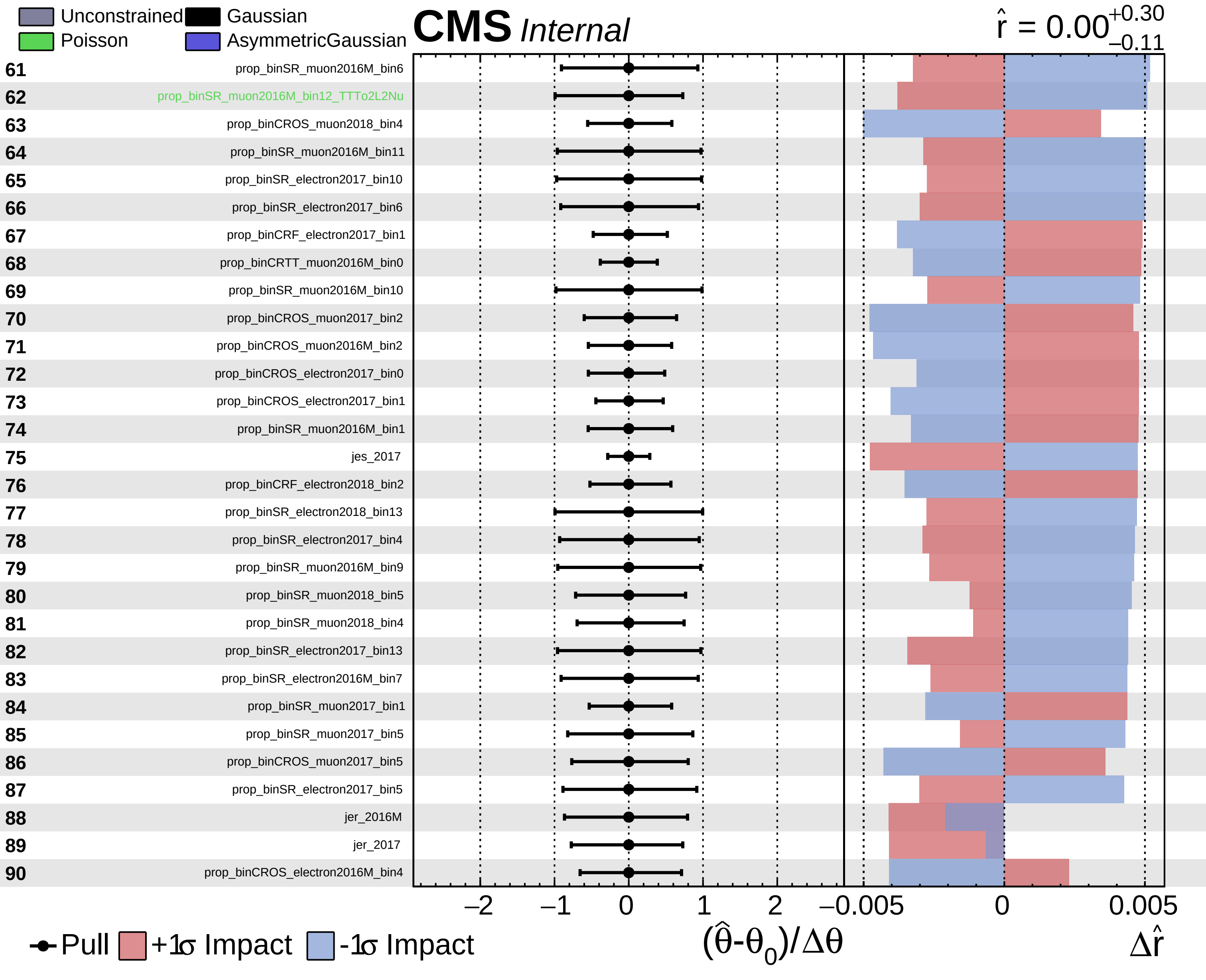


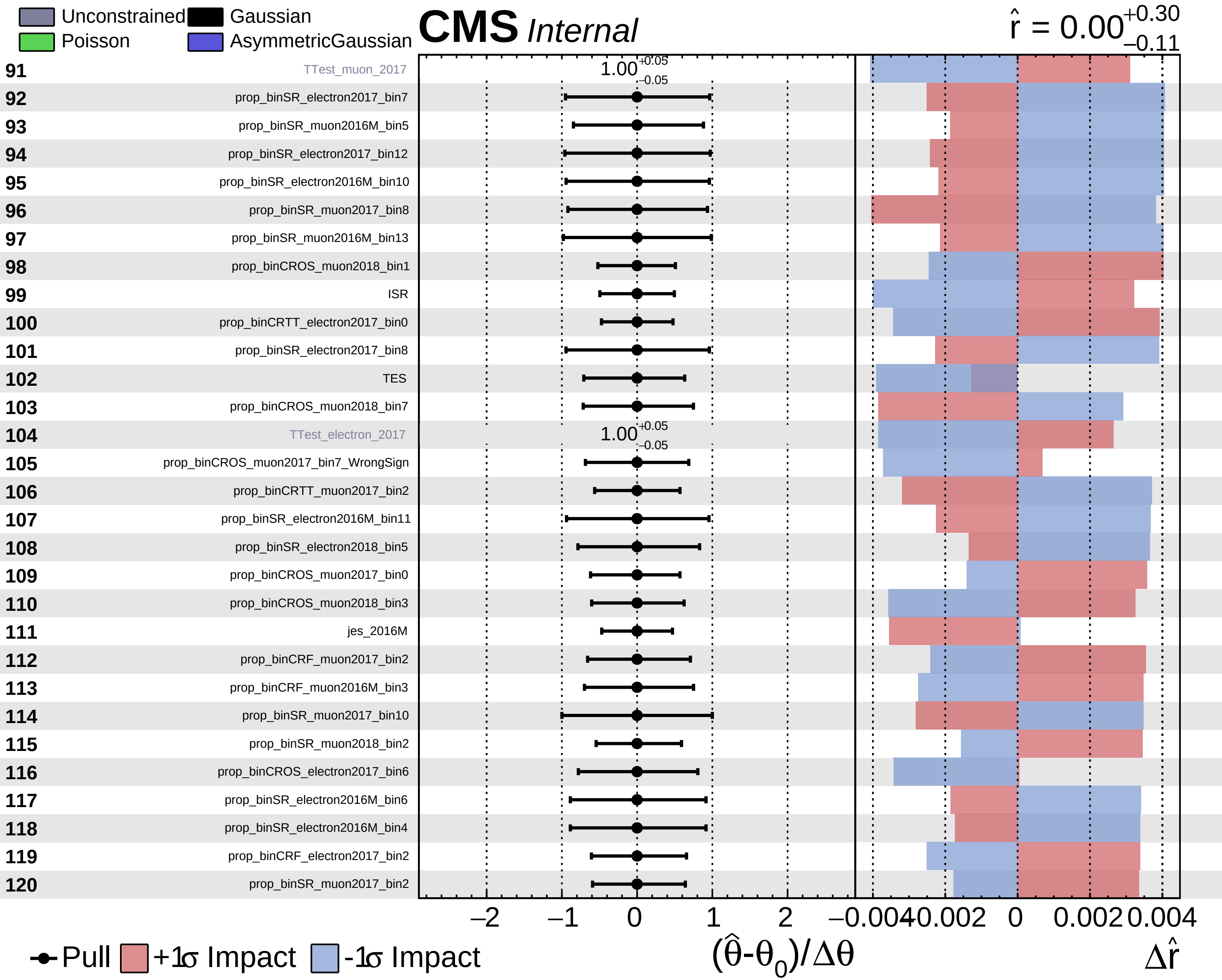
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
1 prop_binCRF_muon2017_bin0
2 FR_sys_muon
3 prop_binCRF_muon2018_bin0
4 prop_binCRF_muon2016M_bin0
5 prop_binSR_muon2017_bin12_VG
6 prop_binCRF_electron2017_bin0
7 prop_binSR_muon2018_bin0
8 FR_sys_electron
9 prop_binCRF_electron2018_bin0
10 prop_binCRF_electron2016M_bin0
11 prop_binSR_muon2017_bin0
12 prop_binCRF_muon2018_bin1
13 prop_binSR_muon2017_bin7_WrongSign
14 OSest_muon_2017
$1.00^{+0.24}_{-0.24}$
15 QCDScale_WpWpJJ_QCD
16 prop_binSR_muon2016M_bin0
17 prop_binSR_muon2018_bin7
18 prop_binSR_muon2018_bin8
19 prop_binSR_muon2018_bin10
20 prop_binSR_muon2018_bin12
21 OSest_muon_2018
$1.00^{+0.19}_{-0.19}$
22 prop_binSR_muon2018_bin11
23 prop_binCRF_muon2016M_bin1
24 OSest_muon_2016M
$1.00^{+0.29}_{-0.29}$
25 prop_binSR_muon2018_bin6
26 metUnclust
27 prop_binCRF_muon2017_bin8_Fake
28 OSest_electron_2017
$1.00^{+0.30}_{-0.30}$
29 prop_binSR_electron2018_bin0
30 prop_binSR_muon2018_bin13
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

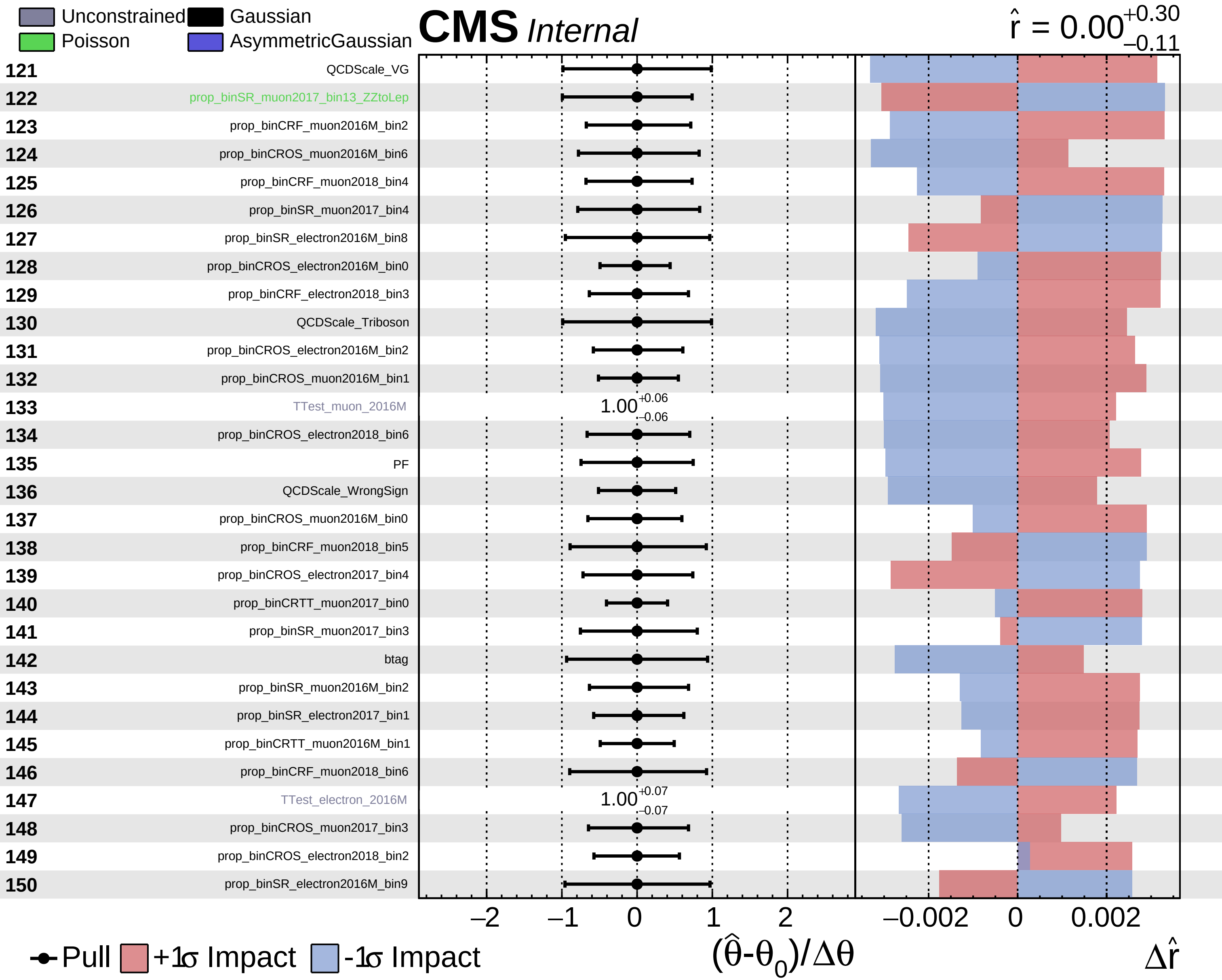


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
31 OSest_electron_2018 $1.00^{+0.29}_{-0.29}$
32 prop_binSR_electron2018_bin12_TTTo2L2Nu
33 prop_binSR_electron2018_bin9
34 pdf_Tot
35 prop_binSR_electron2018_bin11
36 prop_binSR_electron2016M_bin12_WZ
37 prop_binSR_electron2018_bin6
38 prop_binSR_muon2017_bin6
39 prop_binCRF_muon2017_bin1
40 OSest_electron_2016M $1.00^{+0.30}_{-0.30}$
41 prop_binCRF_electron2018_bin1
42 prop_binSR_electron2016M_bin13_Fake
43 prop_binSR_electron2018_bin8
44 prop_binSR_electron2017_bin0
45 prop_binSR_electron2018_bin10
46 prop_binSR_electron2018_bin7
47 prop_binSR_muon2017_bin9
48 prop_binSR_muon2018_bin1
49 prop_binSR_muon2016M_bin8
50 prop_binSR_muon2017_bin11
51 prop_binCROS_electron2018_bin0
52 prop_binCRF_muon2018_bin2
53 prop_binCRF_electron2016M_bin1
54 prop_binSR_electron2016M_bin0
55 FSR
56 prop_binSR_muon2018_bin9
57 QCDScale_TVX
58 jer_2018
59 prop_binSR_electron2017_bin9
60 prop_binSR_muon2016M_bin7
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

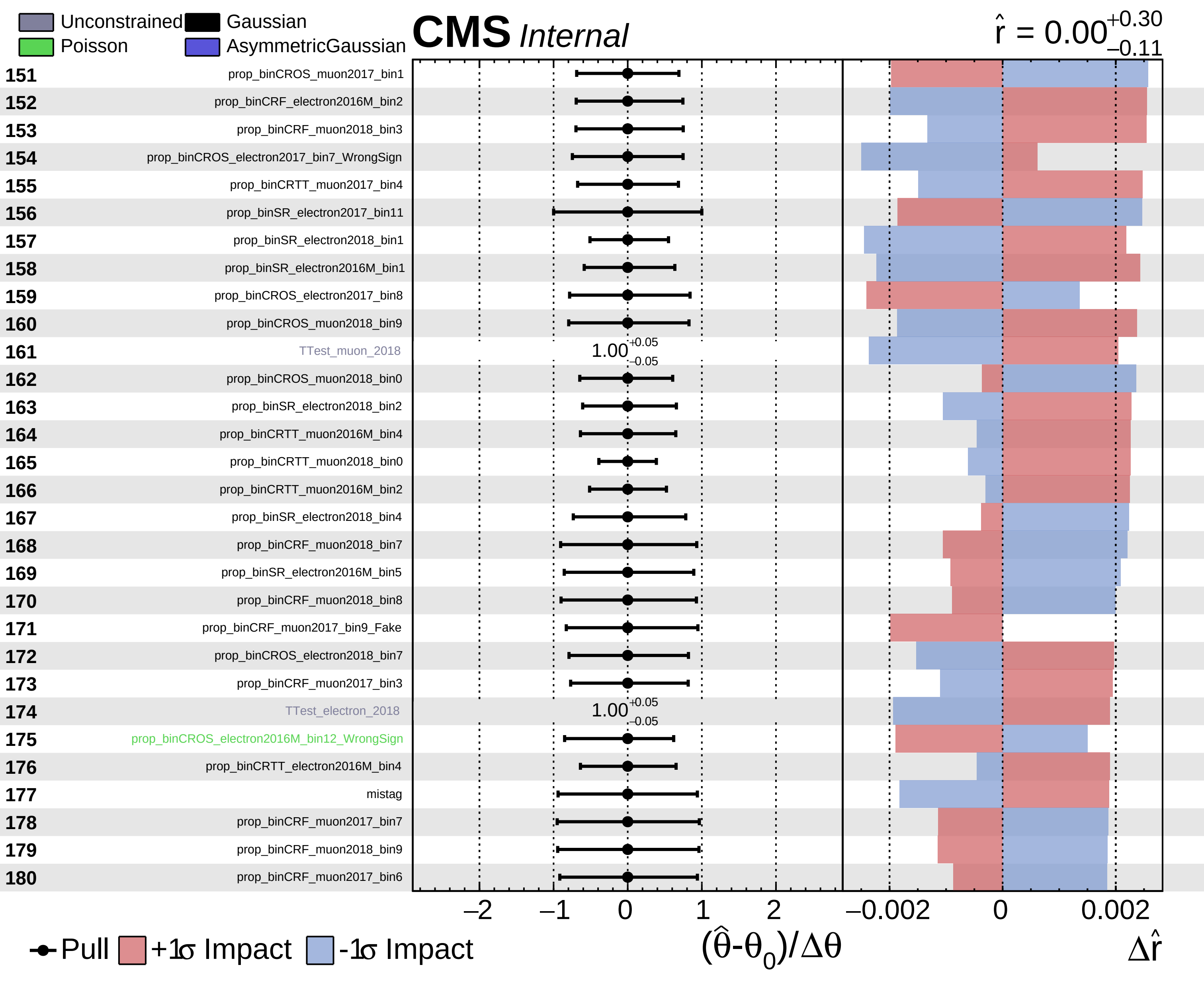


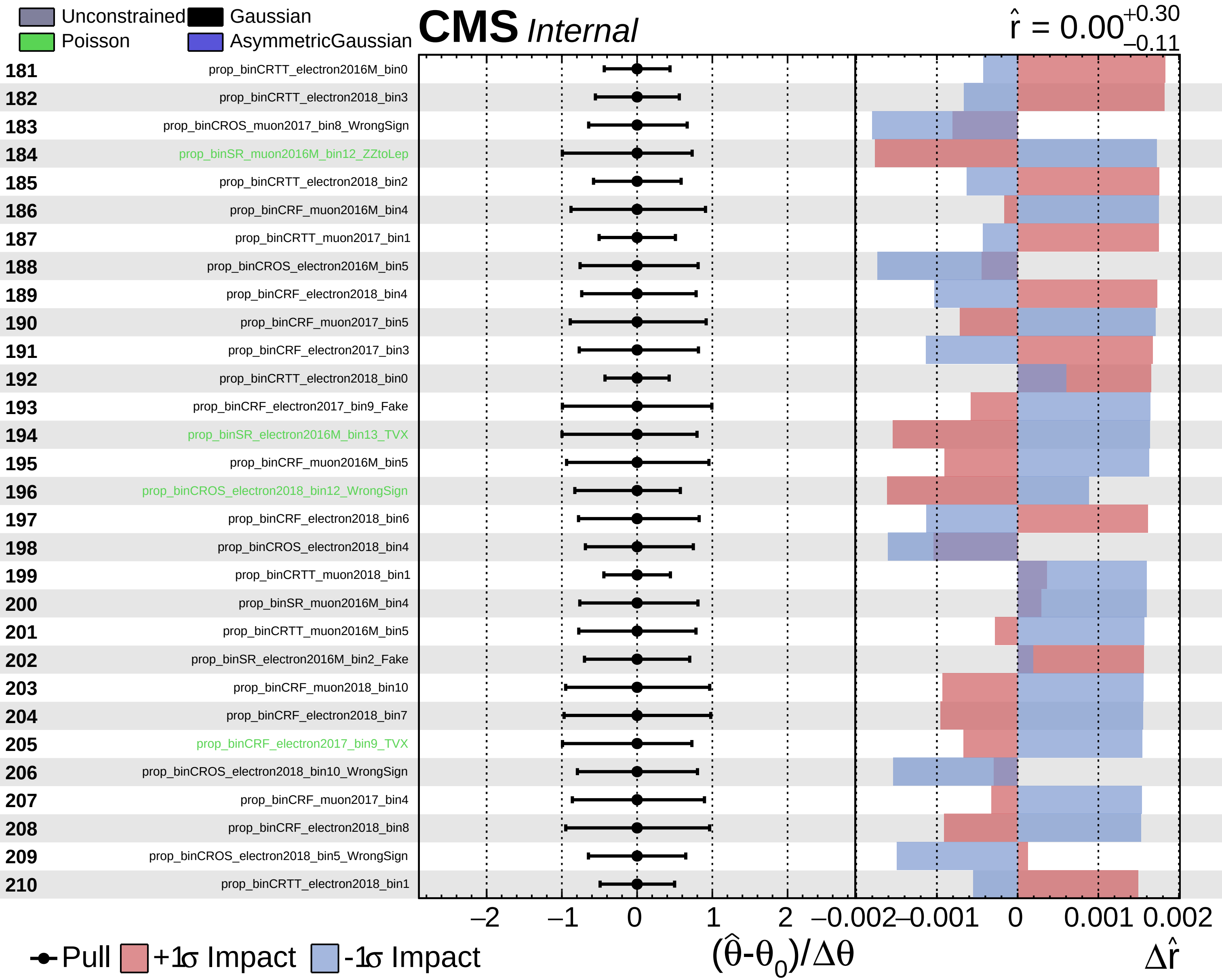
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
61 prop_binSR_muon2016M_bin6
62 prop_binSR_muon2016M_bin12_TTTo2L2Nu
63 prop_binCROS_muon2018_bin4
64 prop_binSR_muon2016M_bin11
65 prop_binSR_electron2017_bin10
66 prop_binSR_electron2017_bin6
67 prop_binCRF_electron2017_bin1
68 prop_binCRTT_muon2016M_bin0
69 prop_binSR_muon2016M_bin10
70 prop_binCROS_muon2017_bin2
71 prop_binCROS_muon2016M_bin2
72 prop_binCROS_electron2017_bin0
73 prop_binCROS_electron2017_bin1
74 prop_binSR_muon2016M_bin1
75 jes_2017
76 prop_binCRF_electron2018_bin2
77 prop_binSR_electron2018_bin13
78 prop_binSR_electron2017_bin4
79 prop_binSR_muon2016M_bin9
80 prop_binSR_muon2018_bin5
81 prop_binSR_muon2018_bin4
82 prop_binSR_electron2017_bin13
83 prop_binSR_electron2016M_bin7
84 prop_binSR_muon2017_bin1
85 prop_binSR_muon2017_bin5
86 prop_binCROS_muon2017_bin5
87 prop_binSR_electron2017_bin5
88 jer_2016M
89 jer_2017
90 prop_binCROS_electron2016M_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
91 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
92 prop_binSR_electron2017_bin7
93 prop_binSR_muon2016M_bin5
94 prop_binSR_electron2017_bin12
95 prop_binSR_electron2016M_bin10
96 prop_binSR_muon2017_bin8
97 prop_binSR_muon2016M_bin13
98 prop_binCROS_muon2018_bin1
99 ISR
100 prop_binCRTT_electron2017_bin0
101 prop_binSR_electron2017_bin8
102 TES
103 prop_binCROS_muon2018_bin7
104 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
105 prop_binCROS_muon2017_bin7_WrongSign
106 prop_binCRTT_muon2017_bin2
107 prop_binSR_electron2016M_bin11
108 prop_binSR_electron2018_bin5
109 prop_binCROS_muon2017_bin0
110 prop_binCROS_muon2018_bin3
111 jes_2016M
112 prop_binCRF_muon2017_bin2
113 prop_binCRF_muon2016M_bin3
114 prop_binSR_muon2017_bin10
115 prop_binSR_muon2018_bin2
116 prop_binCROS_electron2017_bin6
117 prop_binSR_electron2016M_bin6
118 prop_binSR_electron2016M_bin4
119 prop_binCRF_electron2017_bin2
120 prop_binSR_muon2017_bin2
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

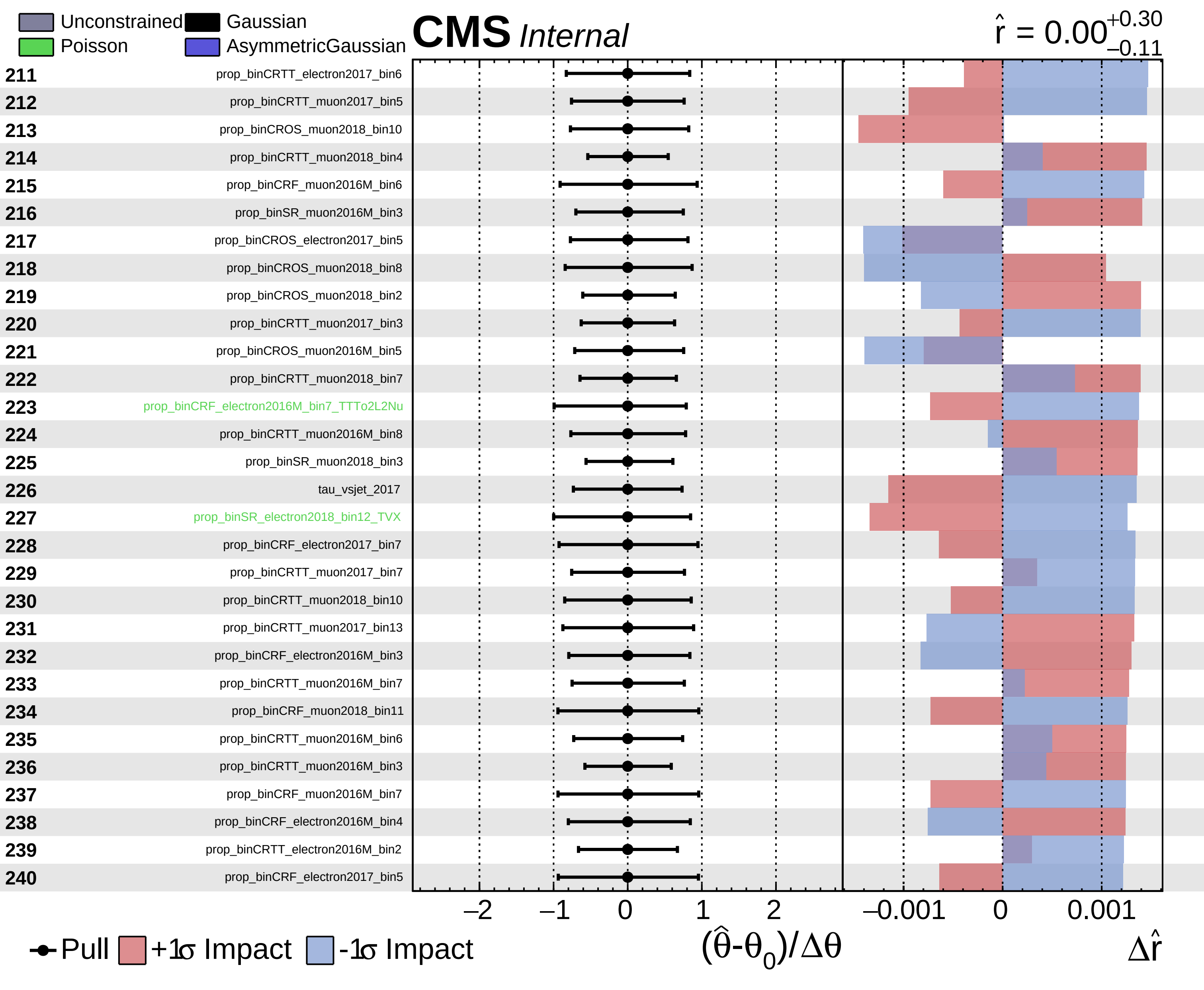


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.30 −0.11
121 QCDScale_VG
122 prop_binSR_muon2017_bin13_ZZtoLep
123 prop_binCRF_muon2016M_bin2
124 prop_binCROS_muon2016M_bin6
125 prop_binCRF_muon2018_bin4
126 prop_binSR_muon2017_bin4
127 prop_binSR_electron2016M_bin8
128 prop_binCROS_electron2016M_bin0
129 prop_binCRF_electron2018_bin3
130 QCDScale_Triboson
131 prop_binCROS_electron2016M_bin2
132 prop_binCROS_muon2016M_bin1
133 TTest_muon_2016M
1.00 +0.06 −0.06
134 prop_binCROS_electron2018_bin6
135 PF
136 QCDScale_WrongSign
137 prop_binCROS_muon2016M_bin0
138 prop_binCRF_muon2018_bin5
139 prop_binCROS_electron2017_bin4
140 prop_binCRTT_muon2017_bin0
141 prop_binSR_muon2017_bin3
142 btag
143 prop_binSR_muon2016M_bin2
144 prop_binSR_electron2017_bin1
145 prop_binCRTT_muon2016M_bin1
146 prop_binCRF_muon2018_bin6
147 TTest_electron_2016M
1.00 +0.07 −0.07
148 prop_binCROS_muon2017_bin3
149 prop_binCROS_electron2018_bin2
150 prop_binSR_electron2016M_bin9
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

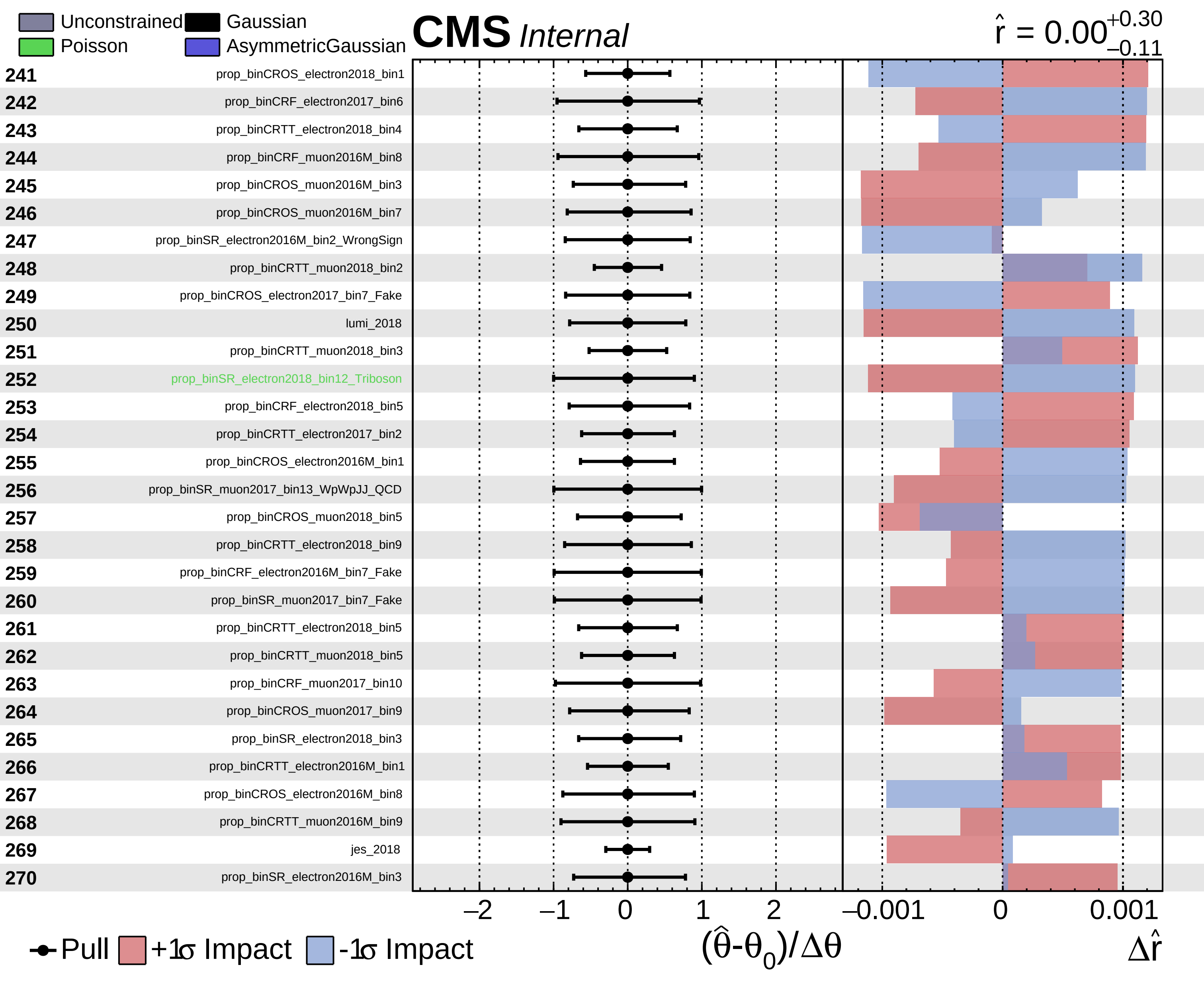

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
151 prop_binCROS_muon2017_bin1
152 prop_binCRF_electron2016M_bin2
153 prop_binCRF_muon2018_bin3
154 prop_binCROS_electron2017_bin7_WrongSign
155 prop_binCRTT_muon2017_bin4
156 prop_binSR_electron2017_bin11
157 prop_binSR_electron2018_bin1
158 prop_binSR_electron2016M_bin1
159 prop_binCROS_electron2017_bin8
160 prop_binCROS_muon2018_bin9
161 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
162 prop_binCROS_muon2018_bin0
163 prop_binSR_electron2018_bin2
164 prop_binCRTT_muon2016M_bin4
165 prop_binCRTT_muon2018_bin0
166 prop_binCRTT_muon2016M_bin2
167 prop_binSR_electron2018_bin4
168 prop_binCRF_muon2018_bin7
169 prop_binSR_electron2016M_bin5
170 prop_binCRF_muon2018_bin8
171 prop_binCRF_muon2017_bin9_Fake
172 prop_binCROS_electron2018_bin7
173 prop_binCRF_muon2017_bin3
174 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
175 prop_binCROS_electron2016M_bin12_WrongSign
176 prop_binCRTT_electron2016M_bin4
177 mistag
178 prop_binCRF_muon2017_bin7
179 prop_binCRF_muon2018_bin9
180 prop_binCRF_muon2017_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



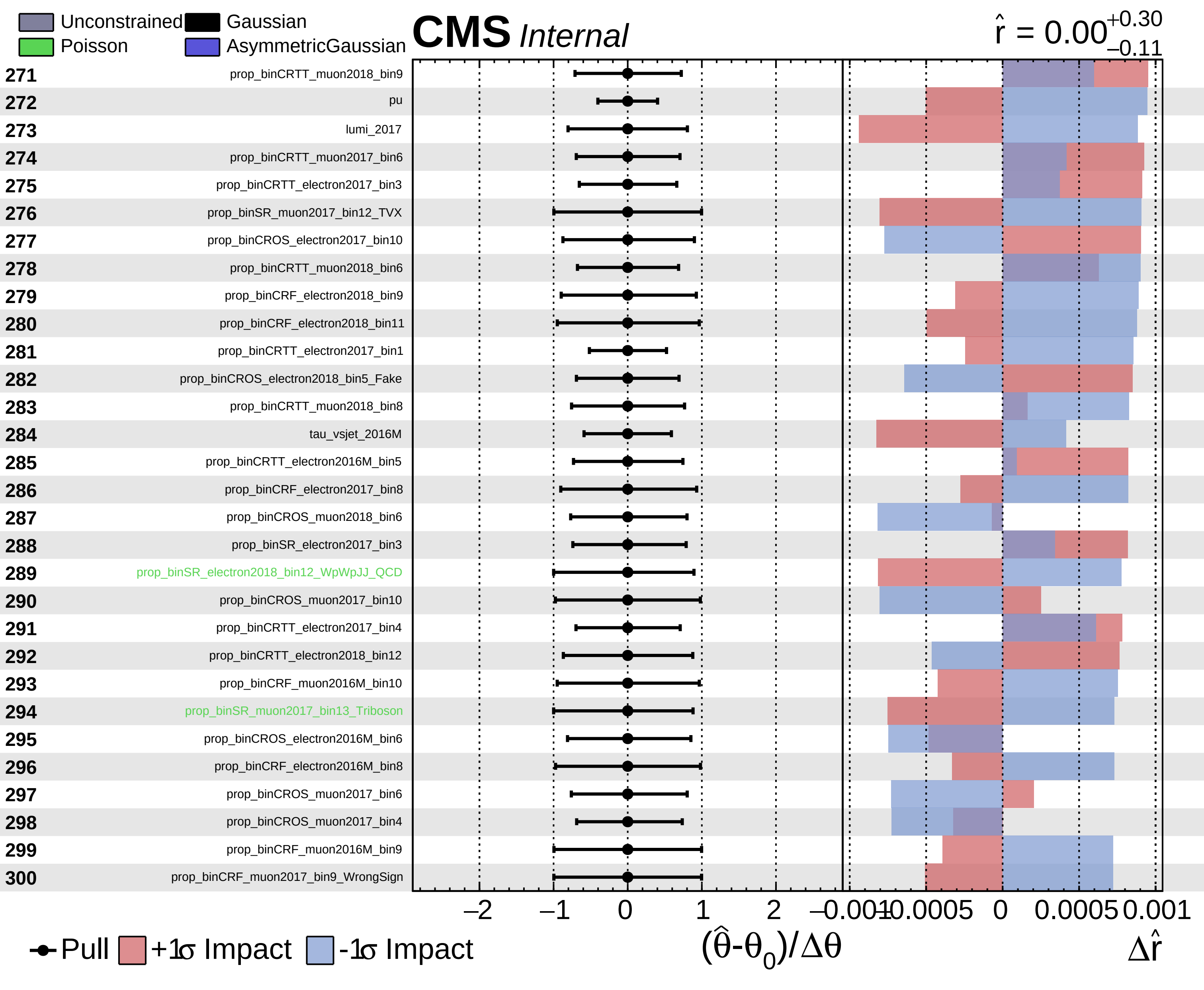
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
181 prop_binCRTT_electron2016M_bin0
182 prop_binCRTT_electron2018_bin3
183 prop_binCROS_muon2017_bin8_WrongSign
184 prop_binSR_muon2016M_bin12_ZZtoLep
185 prop_binCRTT_electron2018_bin2
186 prop_binCRF_muon2016M_bin4
187 prop_binCRTT_muon2017_bin1
188 prop_binCROS_electron2016M_bin5
189 prop_binCRF_electron2018_bin4
190 prop_binCRF_muon2017_bin5
191 prop_binCRF_electron2017_bin3
192 prop_binCRTT_electron2018_bin0
193 prop_binCRF_electron2017_bin9_Fake
194 prop_binSR_electron2016M_bin13_TVX
195 prop_binCRF_muon2016M_bin5
196 prop_binCROS_electron2018_bin12_WrongSign
197 prop_binCRF_electron2018_bin6
198 prop_binCROS_electron2018_bin4
199 prop_binCRTT_muon2018_bin1
200 prop_binSR_muon2016M_bin4
201 prop_binCRTT_muon2016M_bin5
202 prop_binSR_electron2016M_bin2_Fake
203 prop_binCRF_muon2018_bin10
204 prop_binCRF_electron2018_bin7
205 prop_binCRF_electron2017_bin9_TVX
206 prop_binCROS_electron2018_bin10_WrongSign
207 prop_binCRF_muon2017_bin4
208 prop_binCRF_electron2018_bin8
209 prop_binCROS_electron2018_bin5_WrongSign
210 prop_binCRTT_electron2018_bin1
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



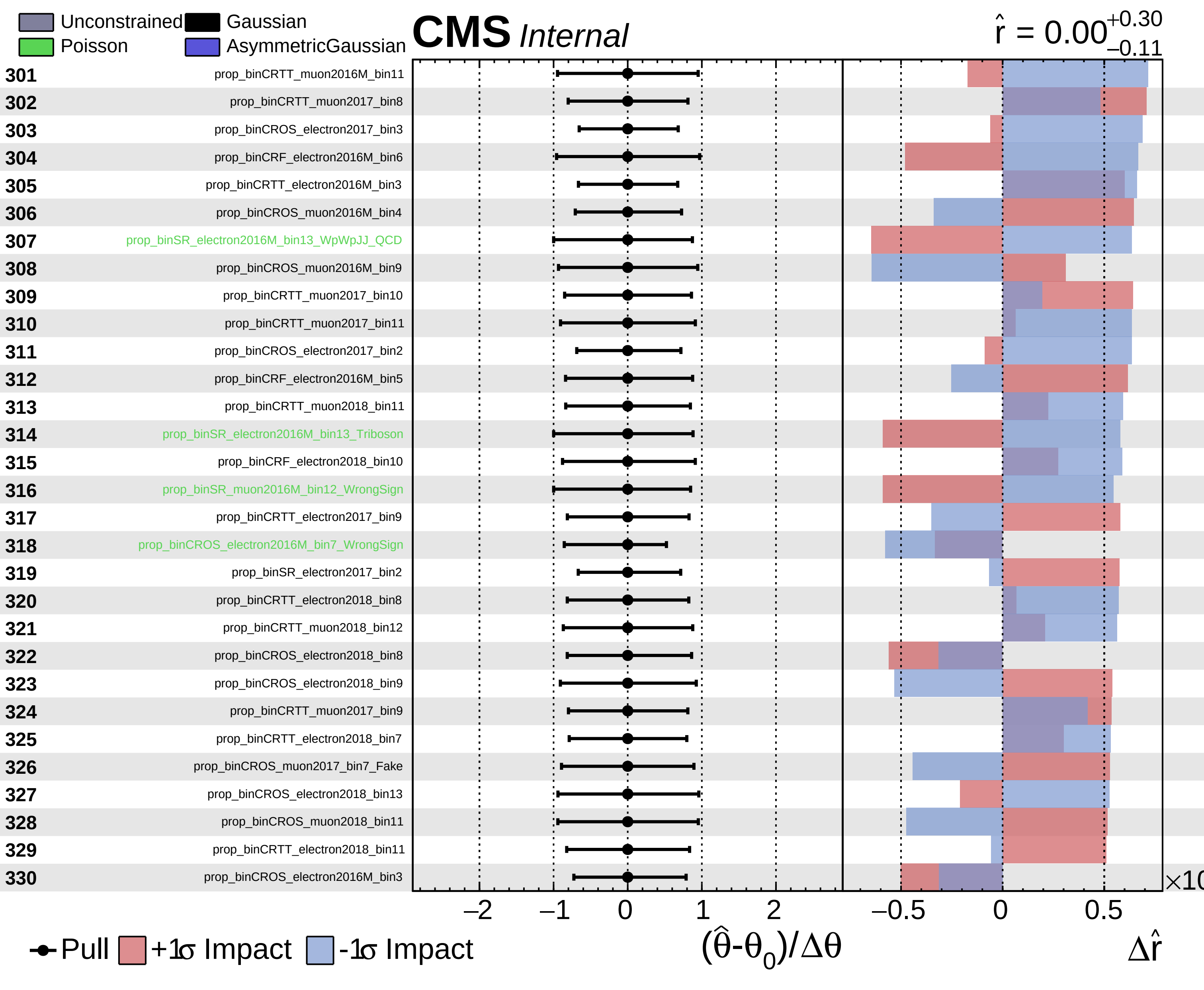
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
211 prop_binCRTT_electron2017_bin6
212 prop_binCRTT_muon2017_bin5
213 prop_binCROS_muon2018_bin10
214 prop_binCRTT_muon2018_bin4
215 prop_binCRF_muon2016M_bin6
216 prop_binSR_muon2016M_bin3
217 prop_binCROS_electron2017_bin5
218 prop_binCROS_muon2018_bin8
219 prop_binCROS_muon2018_bin2
220 prop_binCRTT_muon2017_bin3
221 prop_binCROS_muon2016M_bin5
222 prop_binCRTT_muon2018_bin7
223 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
224 prop_binCRTT_muon2016M_bin8
225 prop_binSR_muon2018_bin3
226 tau_vsjet_2017
227 prop_binSR_electron2018_bin12_TVX
228 prop_binCRF_electron2017_bin7
229 prop_binCRTT_muon2017_bin7
230 prop_binCRTT_muon2018_bin10
231 prop_binCRTT_muon2017_bin13
232 prop_binCRF_electron2016M_bin3
233 prop_binCRTT_muon2016M_bin7
234 prop_binCRF_muon2018_bin11
235 prop_binCRTT_muon2016M_bin6
236 prop_binCRTT_muon2016M_bin3
237 prop_binCRF_muon2016M_bin7
238 prop_binCRF_electron2016M_bin4
239 prop_binCRTT_electron2016M_bin2
240 prop_binCRF_electron2017_bin5
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



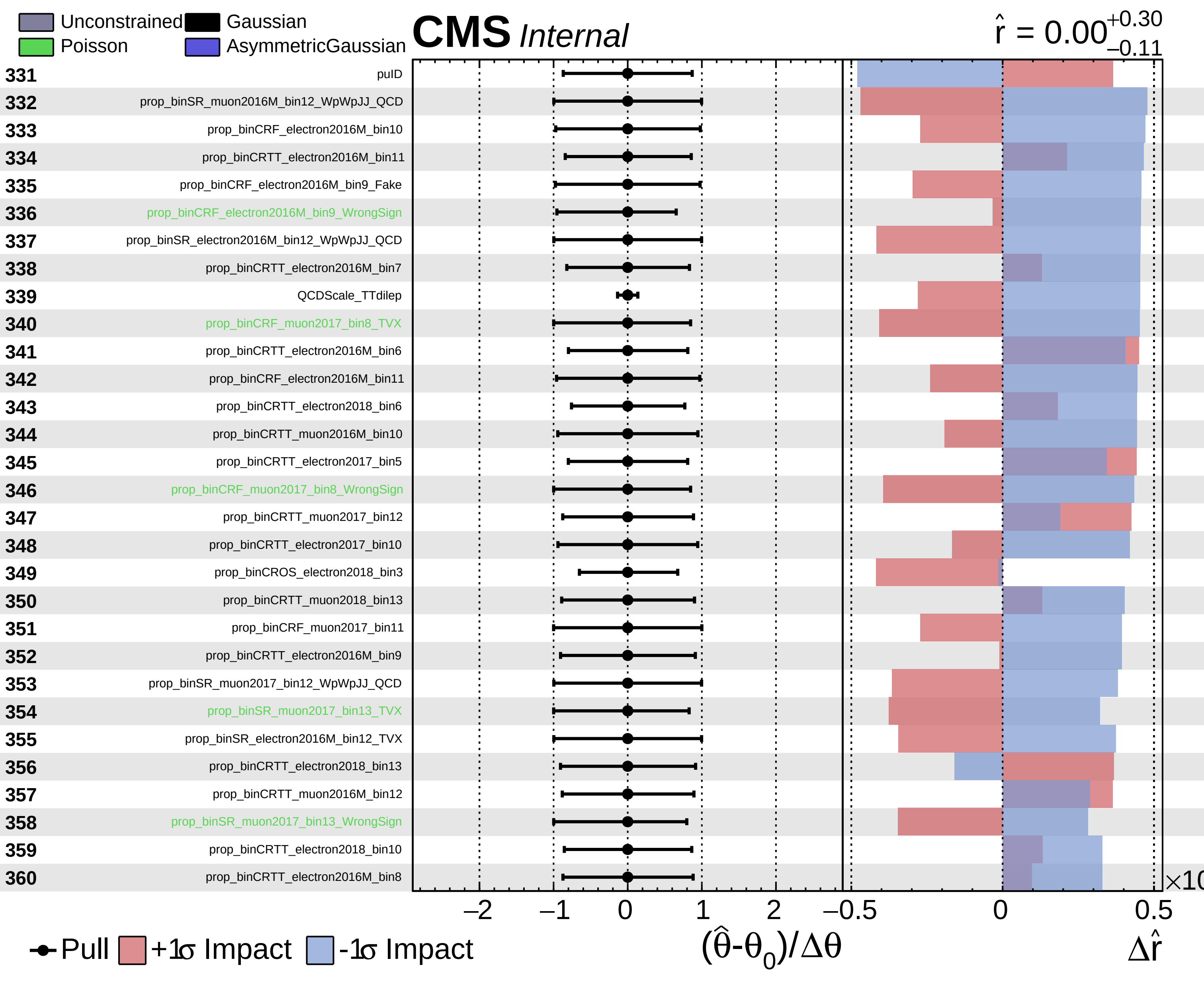
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.30 −0.11
241 prop_binCROS_electron2018_bin1
242 prop_binCRF_electron2017_bin6
243 prop_binCRTT_electron2018_bin4
244 prop_binCRF_muon2016M_bin8
245 prop_binCROS_muon2016M_bin3
246 prop_binCROS_muon2016M_bin7
247 prop_binSR_electron2016M_bin2_WrongSign
248 prop_binCRTT_muon2018_bin2
249 prop_binCROS_electron2017_bin7_Fake
250 lumi_2018
251 prop_binCRTT_muon2018_bin3
252 prop_binSR_electron2018_bin12_Triboson
253 prop_binCRF_electron2018_bin5
254 prop_binCRTT_electron2017_bin2
255 prop_binCROS_electron2016M_bin1
256 prop_binSR_muon2017_bin13_WpWpJJ_QCD
257 prop_binCROS_muon2018_bin5
258 prop_binCRTT_electron2018_bin9
259 prop_binCRF_electron2016M_bin7_Fake
260 prop_binSR_muon2017_bin7_Fake
261 prop_binCRTT_electron2018_bin5
262 prop_binCRTT_muon2018_bin5
263 prop_binCRF_muon2017_bin10
264 prop_binCROS_muon2017_bin9
265 prop_binSR_electron2018_bin3
266 prop_binCRTT_electron2016M_bin1
267 prop_binCROS_electron2016M_bin8
268 prop_binCRTT_muon2016M_bin9
269 jes_2018
270 prop_binSR_electron2016M_bin3
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

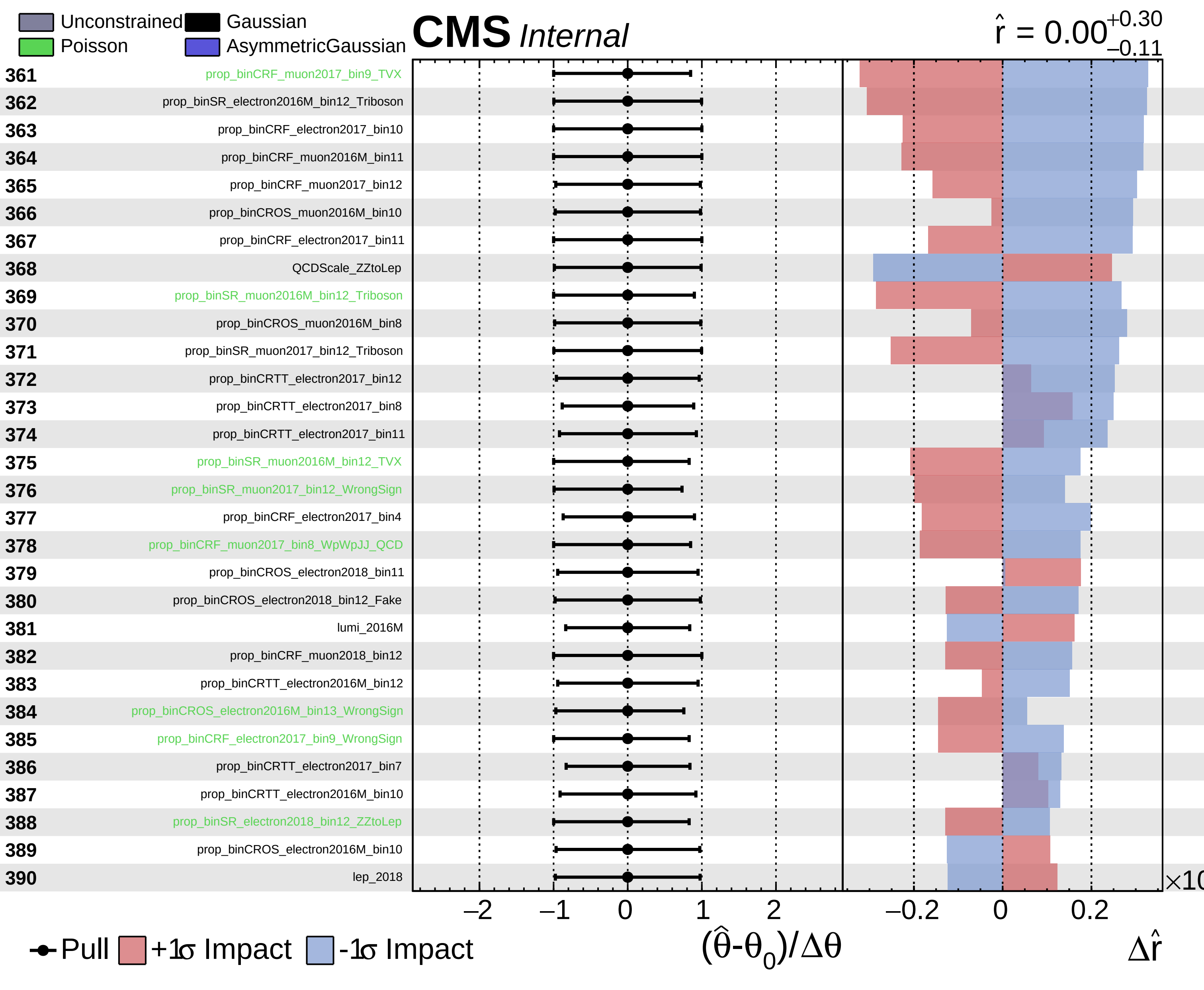


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
271 prop_binCRTT_muon2018_bin9
272 pu
273 lumi_2017
274 prop_binCRTT_muon2017_bin6
275 prop_binCRTT_electron2017_bin3
276 prop_binSR_muon2017_bin12_TVX
277 prop_binCROS_electron2017_bin10
278 prop_binCRTT_muon2018_bin6
279 prop_binCRF_electron2018_bin9
280 prop_binCRF_electron2018_bin11
281 prop_binCRTT_electron2017_bin1
282 prop_binCROS_electron2018_bin5_Fake
283 prop_binCRTT_muon2018_bin8
284 tau_vsjet_2016M
285 prop_binCRTT_electron2016M_bin5
286 prop_binCRF_electron2017_bin8
287 prop_binCROS_muon2018_bin6
288 prop_binSR_electron2017_bin3
289 prop_binSR_electron2018_bin12_WpWpJJ_QCD
290 prop_binCROS_muon2017_bin10
291 prop_binCRTT_electron2017_bin4
292 prop_binCRTT_electron2018_bin12
293 prop_binCRF_muon2016M_bin10
294 prop_binSR_muon2017_bin13_Triboson
295 prop_binCROS_electron2016M_bin6
296 prop_binCRF_electron2016M_bin8
297 prop_binCROS_muon2017_bin6
298 prop_binCROS_muon2017_bin4
299 prop_binCRF_muon2016M_bin9
300 prop_binCRF_muon2017_bin9_WrongSign
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

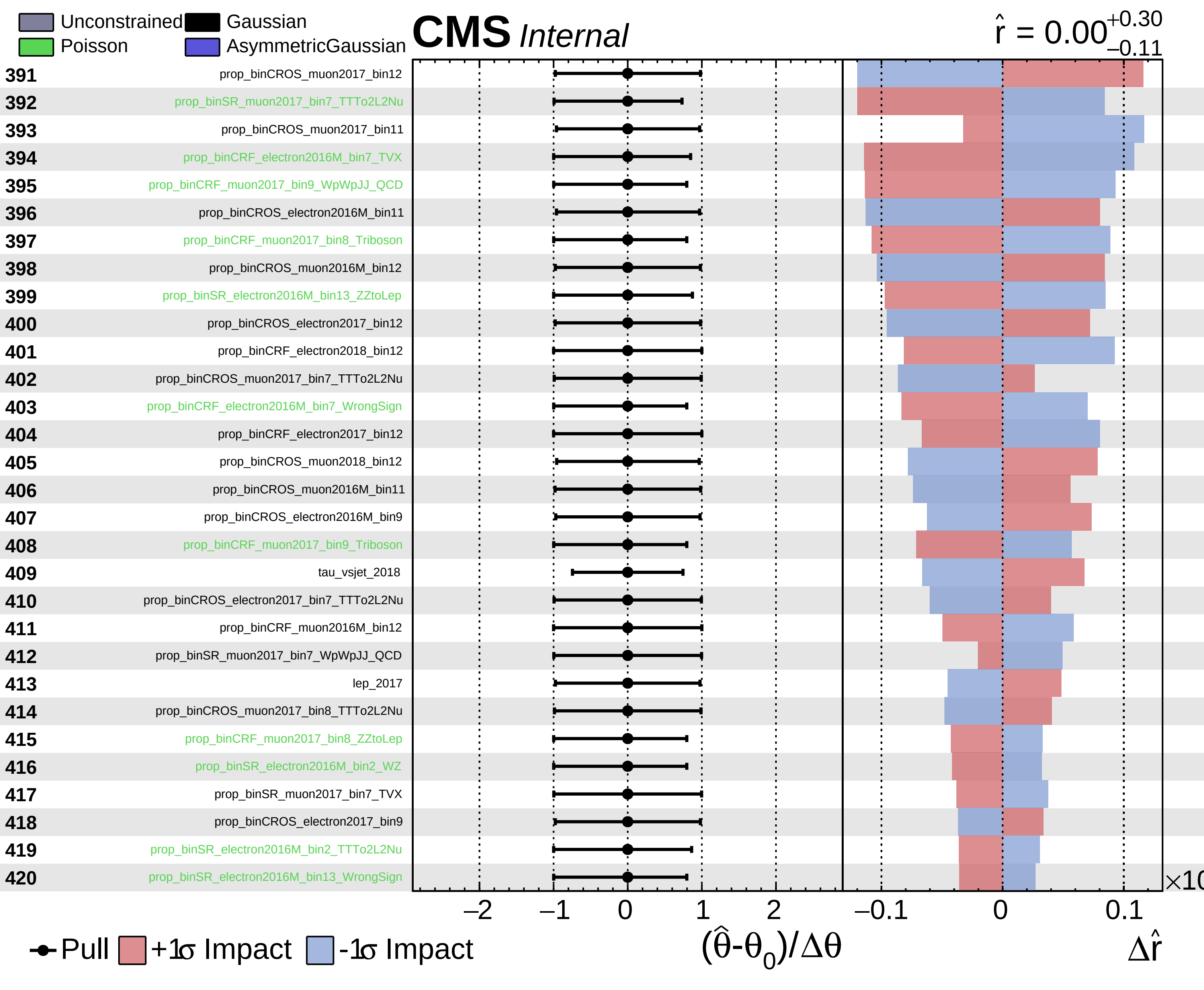


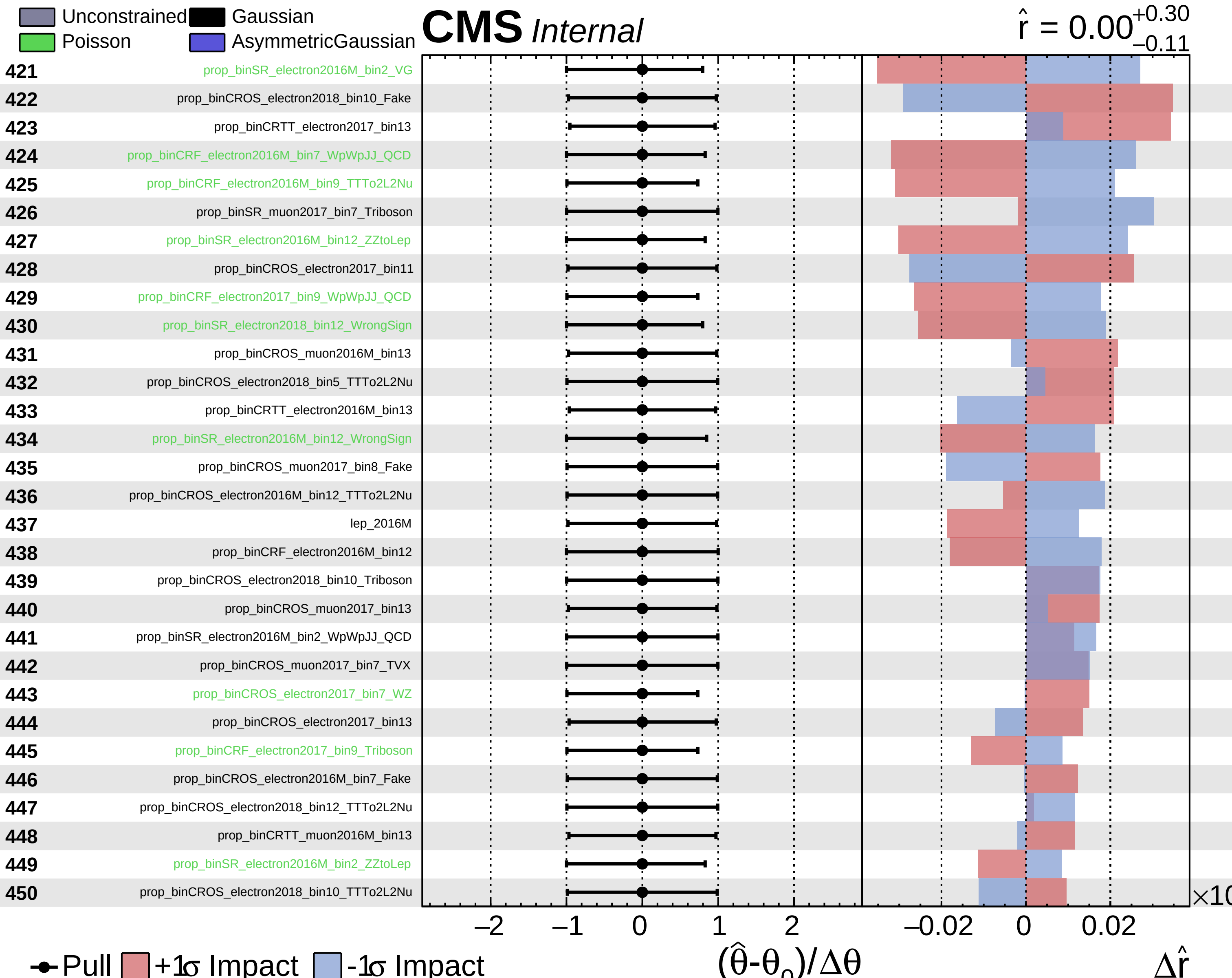
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
301 prop_binCRTT_muon2016M_bin11
302 prop_binCRTT_muon2017_bin8
303 prop_binCROS_electron2017_bin3
304 prop_binCRF_electron2016M_bin6
305 prop_binCRTT_electron2016M_bin3
306 prop_binCROS_muon2016M_bin4
307 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
308 prop_binCROS_muon2016M_bin9
309 prop_binCRTT_muon2017_bin10
310 prop_binCRTT_muon2017_bin11
311 prop_binCROS_electron2017_bin2
312 prop_binCRF_electron2016M_bin5
313 prop_binCRTT_muon2018_bin11
314 prop_binSR_electron2016M_bin13_Triboson
315 prop_binCRF_electron2018_bin10
316 prop_binSR_muon2016M_bin12_WrongSign
317 prop_binCRTT_electron2017_bin9
318 prop_binCROS_electron2016M_bin7_WrongSign
319 prop_binSR_electron2017_bin2
320 prop_binCRTT_electron2018_bin8
321 prop_binCRTT_muon2018_bin12
322 prop_binCROS_electron2018_bin8
323 prop_binCROS_electron2018_bin9
324 prop_binCRTT_muon2017_bin9
325 prop_binCRTT_electron2018_bin7
326 prop_binCROS_muon2017_bin7_Fake
327 prop_binCROS_electron2018_bin13
328 prop_binCROS_muon2018_bin11
329 prop_binCRTT_electron2018_bin11
330 prop_binCROS_electron2016M_bin3
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
331 puID
332 prop_binSR_muon2016M_bin12_WpWpJJ_QCD
333 prop_binCRF_electron2016M_bin10
334 prop_binCRTT_electron2016M_bin11
335 prop_binCRF_electron2016M_bin9_Fake
336 prop_binCRF_electron2016M_bin9_WrongSign
337 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
338 prop_binCRTT_electron2016M_bin7
339 QCDScale_TTdilep
340 prop_binCRF_muon2017_bin8_TVX
341 prop_binCRTT_electron2016M_bin6
342 prop_binCRF_electron2016M_bin11
343 prop_binCRTT_electron2018_bin6
344 prop_binCRTT_muon2016M_bin10
345 prop_binCRTT_electron2017_bin5
346 prop_binCRF_muon2017_bin8_WrongSign
347 prop_binCRTT_muon2017_bin12
348 prop_binCRTT_electron2017_bin10
349 prop_binCROS_electron2018_bin3
350 prop_binCRTT_muon2018_bin13
351 prop_binCRF_muon2017_bin11
352 prop_binCRTT_electron2016M_bin9
353 prop_binSR_muon2017_bin12_WpWpJJ_QCD
354 prop_binSR_muon2017_bin13_TVX
355 prop_binSR_electron2016M_bin12_TVX
356 prop_binCRTT_electron2018_bin13
357 prop_binCRTT_muon2016M_bin12
358 prop_binSR_muon2017_bin13_WrongSign
359 prop_binCRTT_electron2018_bin10
360 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

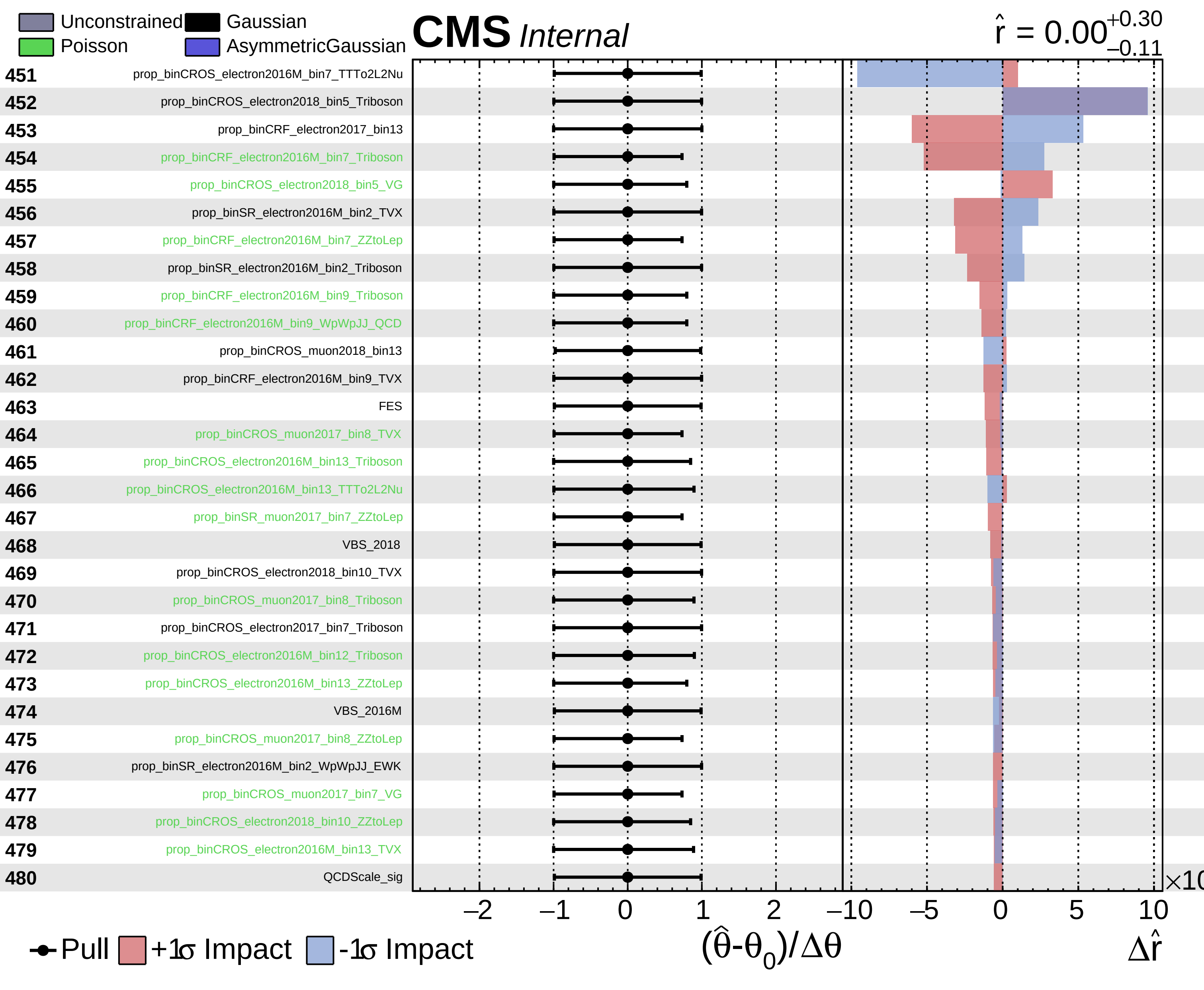


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
361 prop_binCRF_muon2017_bin9_TVX
362 prop_binSR_electron2016M_bin12_Triboson
363 prop_binCRF_electron2017_bin10
364 prop_binCRF_muon2016M_bin11
365 prop_binCRF_muon2017_bin12
366 prop_binCROS_muon2016M_bin10
367 prop_binCRF_electron2017_bin11
368 QCDScale_ZZtoLep
369 prop_binSR_muon2016M_bin12_Triboson
370 prop_binCROS_muon2016M_bin8
371 prop_binSR_muon2017_bin12_Triboson
372 prop_binCRTT_electron2017_bin12
373 prop_binCRTT_electron2017_bin8
374 prop_binCRTT_electron2017_bin11
375 prop_binSR_muon2016M_bin12_TVX
376 prop_binSR_muon2017_bin12_WrongSign
377 prop_binCRF_electron2017_bin4
378 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
379 prop_binCROS_electron2018_bin11
380 prop_binCROS_electron2018_bin12_Fake
381 lumi_2016M
382 prop_binCRF_muon2018_bin12
383 prop_binCRTT_electron2016M_bin12
384 prop_binCROS_electron2016M_bin13_WrongSign
385 prop_binCRF_electron2017_bin9_WrongSign
386 prop_binCRTT_electron2017_bin7
387 prop_binCRTT_electron2016M_bin10
388 prop_binSR_electron2018_bin12_ZZtoLep
389 prop_binCROS_electron2016M_bin10
390 lep_2018
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

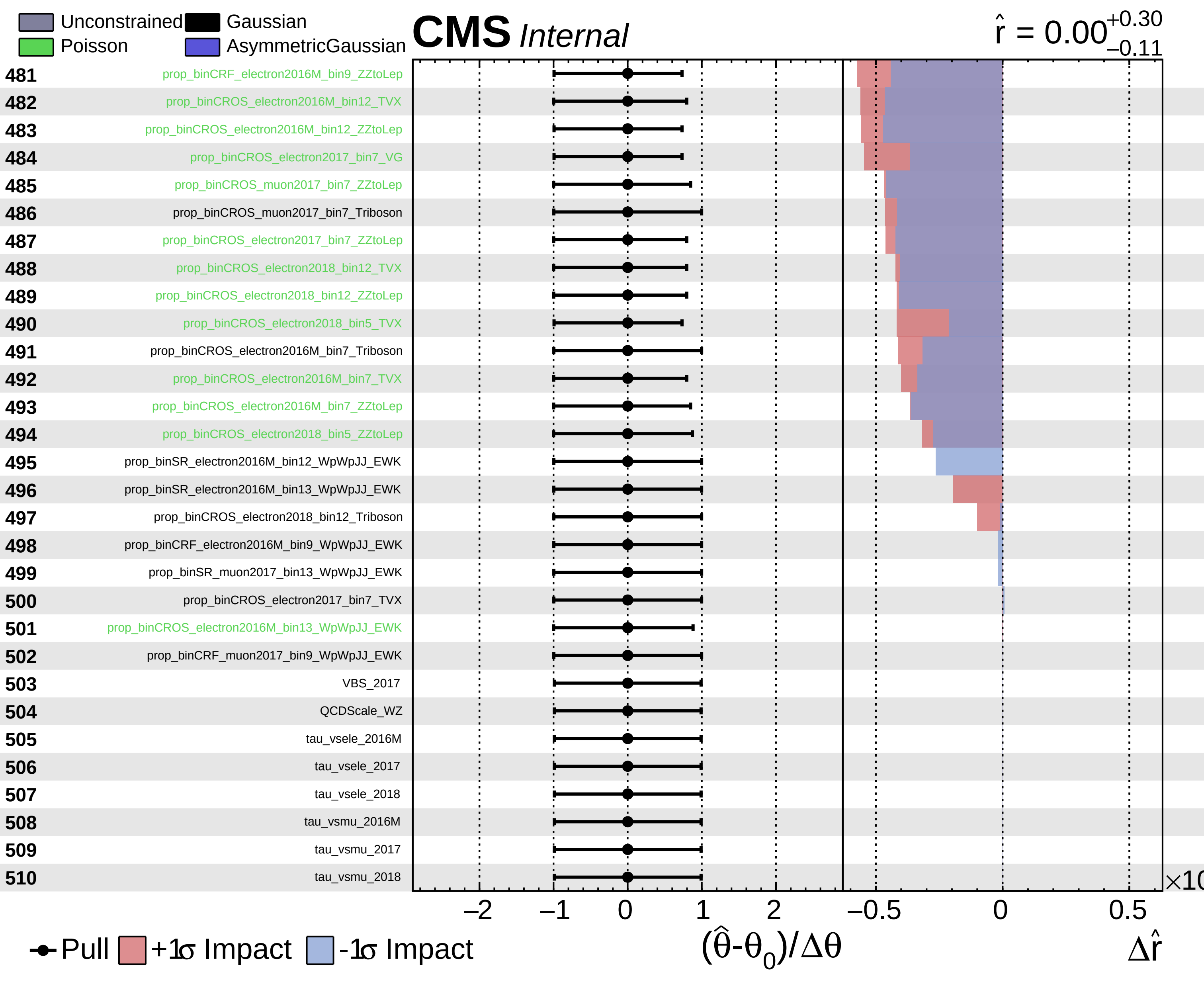

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
391 prop_binCROS_muon2017_bin12
392 prop_binSR_muon2017_bin7_TTTo2L2Nu
393 prop_binCROS_muon2017_bin11
394 prop_binCRF_electron2016M_bin7_TVX
395 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
396 prop_binCROS_electron2016M_bin11
397 prop_binCRF_muon2017_bin8_Triboson
398 prop_binCROS_muon2016M_bin12
399 prop_binSR_electron2016M_bin13_ZZtoLep
400 prop_binCROS_electron2017_bin12
401 prop_binCRF_electron2018_bin12
402 prop_binCROS_muon2017_bin7_TTTo2L2Nu
403 prop_binCRF_electron2016M_bin7_WrongSign
404 prop_binCRF_electron2017_bin12
405 prop_binCROS_muon2018_bin12
406 prop_binCROS_muon2016M_bin11
407 prop_binCROS_electron2016M_bin9
408 prop_binCRF_muon2017_bin9_Triboson
409 tau_vsjet_2018
410 prop_binCROS_electron2017_bin7_TTTo2L2Nu
411 prop_binCRF_muon2016M_bin12
412 prop_binSR_muon2017_bin7_WpWpJJ_QCD
413 lep_2017
414 prop_binCROS_muon2017_bin8_TTTo2L2Nu
415 prop_binCRF_muon2017_bin8_ZZtoLep
416 prop_binSR_electron2016M_bin2_WZ
417 prop_binSR_muon2017_bin7_TVX
418 prop_binCROS_electron2017_bin9
419 prop_binSR_electron2016M_bin2_TTTo2L2Nu
420 prop_binSR_electron2016M_bin13_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



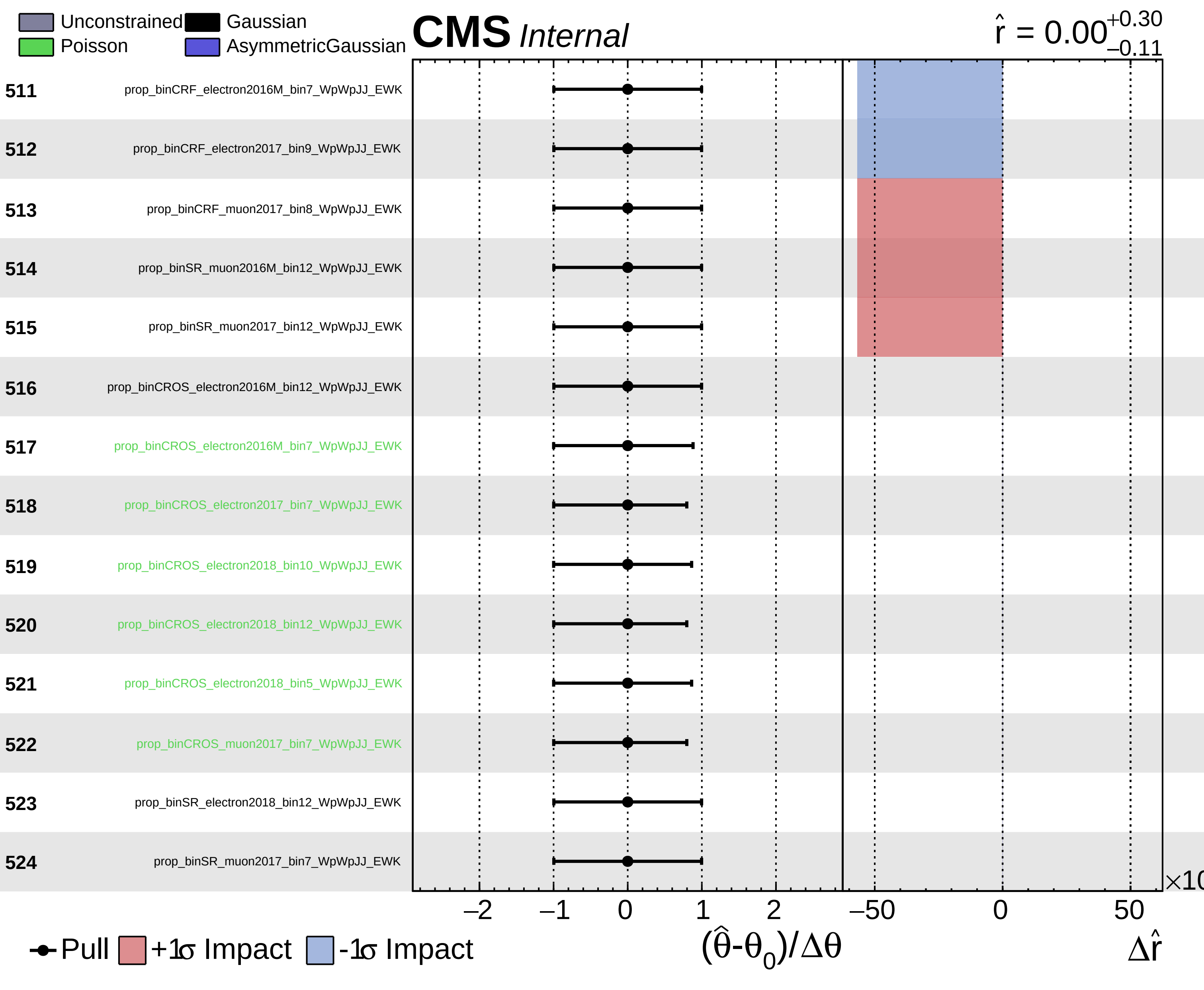
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
421 prop_binSR_electron2016M_bin2_VG
422 prop_binCROS_electron2018_bin10_Fake
423 prop_binCRTT_electron2017_bin13
424 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
425 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
426 prop_binSR_muon2017_bin7_Triboson
427 prop_binSR_electron2016M_bin12_ZZtoLep
428 prop_binCROS_electron2017_bin11
429 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
430 prop_binSR_electron2018_bin12_WrongSign
431 prop_binCROS_muon2016M_bin13
432 prop_binCROS_electron2018_bin5_TTTo2L2Nu
433 prop_binCRTT_electron2016M_bin13
434 prop_binSR_electron2016M_bin12_WrongSign
435 prop_binCROS_muon2017_bin8_Fake
436 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
437 lep_2016M
438 prop_binCRF_electron2016M_bin12
439 prop_binCROS_electron2018_bin10_Triboson
440 prop_binCROS_muon2017_bin13
441 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
442 prop_binCROS_muon2017_bin7_TVX
443 prop_binCROS_electron2017_bin7_WZ
444 prop_binCROS_electron2017_bin13
445 prop_binCRF_electron2017_bin9_Triboson
446 prop_binCROS_electron2016M_bin7_Fake
447 prop_binCROS_electron2018_bin12_TTTo2L2Nu
448 prop_binCRTT_muon2016M_bin13
449 prop_binSR_electron2016M_bin2_ZZtoLep
450 prop_binCROS_electron2018_bin10_TTTo2L2Nu
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
451 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
452 prop_binCROS_electron2018_bin5_Triboson
453 prop_binCRF_electron2017_bin13
454 prop_binCRF_electron2016M_bin7_Triboson
455 prop_binCROS_electron2018_bin5_VG
456 prop_binSR_electron2016M_bin2_TVX
457 prop_binCRF_electron2016M_bin7_ZZtoLep
458 prop_binSR_electron2016M_bin2_Triboson
459 prop_binCRF_electron2016M_bin9_Triboson
460 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
461 prop_binCROS_muon2018_bin13
462 prop_binCRF_electron2016M_bin9_TVX
463 FES
464 prop_binCROS_muon2017_bin8_TVX
465 prop_binCROS_electron2016M_bin13_Triboson
466 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
467 prop_binSR_muon2017_bin7_ZZtoLep
468 VBS_2018
469 prop_binCROS_electron2018_bin10_TVX
470 prop_binCROS_muon2017_bin8_Triboson
471 prop_binCROS_electron2017_bin7_Triboson
472 prop_binCROS_electron2016M_bin12_Triboson
473 prop_binCROS_electron2016M_bin13_ZZtoLep
474 VBS_2016M
475 prop_binCROS_muon2017_bin8_ZZtoLep
476 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
477 prop_binCROS_muon2017_bin7_VG
478 prop_binCROS_electron2018_bin10_ZZtoLep
479 prop_binCROS_electron2016M_bin13_TVX
480 QCDScale_sig
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.30 −0.11
481 prop_binCRF_electron2016M_bin9_ZZtoLep
482 prop_binCROS_electron2016M_bin12_TVX
483 prop_binCROS_electron2016M_bin12_ZZtoLep
484 prop_binCROS_electron2017_bin7_VG
485 prop_binCROS_muon2017_bin7_ZZtoLep
486 prop_binCROS_muon2017_bin7_Triboson
487 prop_binCROS_electron2017_bin7_ZZtoLep
488 prop_binCROS_electron2018_bin12_TVX
489 prop_binCROS_electron2018_bin12_ZZtoLep
490 prop_binCROS_electron2018_bin5_TVX
491 prop_binCROS_electron2016M_bin7_Triboson
492 prop_binCROS_electron2016M_bin7_TVX
493 prop_binCROS_electron2016M_bin7_ZZtoLep
494 prop_binCROS_electron2018_bin5_ZZtoLep
495 prop_binSR_electron2016M_bin12_WpWpJJ_EWK
496 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
497 prop_binCROS_electron2018_bin12_Triboson
498 prop_binCRF_electron2016M_bin9_WpWpJJ_EWK
499 prop_binSR_muon2017_bin13_WpWpJJ_EWK
500 prop_binCROS_electron2017_bin7_TVX
501 prop_binCROS_electron2016M_bin13_WpWpJJ_EWK
502 prop_binCRF_muon2017_bin9_WpWpJJ_EWK
503 VBS_2017
504 QCDScale_WZ
505 tau_vsele_2016M
506 tau_vsele_2017
507 tau_vsele_2018
508 tau_vsmu_2016M
509 tau_vsmu_2017
510 tau_vsmu_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.30}_{-0.11}$
511 prop_binCRF_electron2016M_bin7_WpWpJJ_EWK
512 prop_binCRF_electron2017_bin9_WpWpJJ_EWK
513 prop_binCRF_muon2017_bin8_WpWpJJ_EWK
514 prop_binSR_muon2016M_bin12_WpWpJJ_EWK
515 prop_binSR_muon2017_bin12_WpWpJJ_EWK
516 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
517 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
518 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
519 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
520 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
521 prop_binCROS_electron2018_bin5_WpWpJJ_EWK
522 prop_binCROS_muon2017_bin7_WpWpJJ_EWK
523 prop_binSR_electron2018_bin12_WpWpJJ_EWK
524 prop_binSR_muon2017_bin7_WpWpJJ_EWK
−2 −1 0 1 2
−50 0 50
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$